

A planar tight signed graph

Disproof of conjecture 00000002161

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Abstract

The all-positive four-vertex path has independence number 2, positive inertia index 2, and minimum orientable genus 0. It is connected and noncomplete. It therefore contradicts the asserted minimum genus 1, whether the minimum is taken separately for every eligible graph or across the eligible class. A Lean 4 project verifies the counterexample using the standard rotation-system definition of orientable cellular embeddings.

1 Statement and scope

The exact bilingual source is supplied as `conjecture.md`. Its English text is:

Definition: The inertia bound of signed graphs states the independence number is at most the positive inertia index. Conjecture: The minimal genus of graphs attaining the tight bound equals 1 (apart from complete graphs); the genus is computed explicitly via the Euler formula of 2-cell embeddings.

We retain the printed equality condition $\alpha(G) = i_+(A)$, where A is the signed adjacency matrix. We neither replace the printed inequality by a conventional inertia bound nor assume it holds for every signed graph. The witness satisfies the printed equality itself. Genus has its usual orientable meaning; a two-cell embedding has open disks as complementary faces. This submission makes no claim about a reformulation requiring a strictly nonorientable surface.

2 A tight, connected, noncomplete witness

Take the path $0-1-2-3$, with all three edge signs $+1$. The vertices $0, 2$ are nonadjacent, so the graph is not complete. The path is connected. The set $\{0, 2\}$ is independent, while the disjoint edges 01 and 23 cover all vertices and each contributes at most one member to an independent set. Thus $\alpha(P_4) = 2$.

The adjacency matrix and its characteristic polynomial are

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad p(t) = \det(tI - A) = t^4 - 3t^2 + 1.$$

Put $r = (\sqrt{5} + 1)/2$ and $s = (\sqrt{5} - 1)/2$. Both are positive, $rs = 1$, and $r^2 + s^2 = 3$. Hence

$$p(t) = (t - r)(t - s)(t + r)(t + s).$$

Its complete root multiset is $[r, s, -r, -s]$. Counting positive eigenvalues with algebraic multiplicity gives $i_+(A) = 2$; no numerical approximation is used. This is the ordinary inertia convention for real symmetric matrices [2].

3 An actual cellular embedding and its genus

3.1 Rotation systems and their meaning

For a connected finite graph with no isolated vertex, use one dart for each ordered adjacent pair. Let λ reverse darts and let ρ cyclically order the darts with each fixed tail. The formal data require that reversal is a fixed-point-free involution, that the dart-to-ordered-edge map is bijective, and that the rotation preserves tails and has one orbit in each vertex star. These conditions describe the original graph, not a different multigraph or an arbitrary pair of permutations.

This is the standard combinatorial encoding of an orientable cellular embedding [1, Section 2, Proposition 2.2]. Its geometric meaning can be seen by taking oriented vertex disks, joining their marked boundary intervals with untwisted edge bands, and capping the resulting boundary circles with disks. The boundary components correspond to cycles of $\rho\lambda$, with λ applied first. Conversely, an oriented cellular embedding gives the local cyclic orders.

For this path specifically, begin with the vertex-0 disk and attach each successive band and new vertex disk. Each step joins two disks along a band and produces a disk. The thickening is therefore a disk; capping its one boundary circle gives the sphere. The central edge arcs have the prescribed endpoints and disjoint interiors. Thus a genuine genus-zero cellular embedding exists, independently of merely guessing a genus from numbers.

3.2 The complete finite certificate

The following table fixes all dart data; entries in the last two rows are dart labels.

Dart d	0	1	2	3	4	5
Directed edge	01	10	12	21	23	32
$\lambda(d)$	1	0	3	2	5	4
$\rho(d)$	0	2	1	4	3	5

Thus $\lambda = (0\ 1)(2\ 3)(4\ 5)$, $\rho = (0)(1\ 2)(3\ 4)(5)$, and

$$\rho\lambda = (0\ 2\ 4\ 5\ 3\ 1).$$

There is one face, with boundary walk $0, 1, 2, 3, 2, 1, 0$. Repeated edges in this boundary are allowed: each bridge borders the same face on both sides. The vertex, edge, and face counts are $V = 4$, $E = 3$, $F = 1$. Euler's formula for this orientable cellular embedding gives

$$2 - 2g = V - E + F = 4 - 3 + 1 = 2, \quad g = 0.$$

The minimum genus is also 0: it is attained here and every orientable genus is a nonnegative integer.

Counterexample. *The signed path P_4 is connected and noncomplete, attains $\alpha(P_4) = i_+(A) = 2$, and has minimum genus 0.*

If the conjecture asserts minimum genus 1 for every tight noncomplete graph, this witness contradicts it directly. If it asserts a class minimum of 1, it necessarily asserts a lower bound 1 for each eligible graph's minimum genus; the same witness contradicts that clause. Restricting the formal universal clauses to connected graphs is sufficient, since a counterexample in that subclass is a counterexample in the full class.

4 Formal specification and coverage

The project uses Lean 4.19.0 and Mathlib v4.19.0, with the exact dependency revisions recorded in `lean/lake-manifest.json`. All statements below are in namespace `Conjecture2161`.

Definition or theorem	What is represented or proved
<code>SignedGraph</code>	An actual simple graph, symmetric real matrix, edge entries ± 1 , and zero entries on every nonedge.
<code>independenceNumber</code>	The maximum cardinality over all finite independent subsets, with its generic upper and lower specifications proved.
<code>path_independence_number</code>	The witness set and upper bound over all 16 subsets establish the exact value 2.
<code>path_roots</code> , <code>path_positive_inertia</code>	The actual matrix characteristic polynomial has the complete root multiset above; its positive part has cardinality 2.
<code>RotationSystem</code>	Every incidence, reversal, graph-connectedness, nonempty-star and cyclic-order condition described on page 2.
<code>RotationSystem.Face</code>	The quotient of darts by the actual face-permutation orbit relation. A named instance proves this type finite.
<code>HasCellularGenus</code>	Existence of a valid rotation system satisfying $V - E + F = 2 - 2g$ with integer arithmetic.
<code>IsMinimumGenus</code>	A realized genus and a lower bound for every other realized genus; no empty infimum or default genus value.
<code>counterexample_certificate</code>	Connectedness, noncompleteness, exact printed equality, and minimum genus 0.
<code>not_per_graph_genus_one</code> , <code>not_class_minimum_one</code>	Negations of the necessary clauses for the two readings of the conjecture.

Formalization boundary. Lean checks the entire finite rotation-system certificate, actual face quotient cardinality, integer Euler equality, existential genus certificate, minimum argument, and final negations. The project uses the standard combinatorial definition of orientable cellular embedding, with the geometric correspondence explained above and cited. It does not additionally formalize the disk-and-band realization theorem in Mathlib’s general topology language. No geometric-realization axiom is inserted into Lean. The restriction to connected noncomplete graphs avoids the isolated-vertex degeneracy of a dart-only representation.

Trust and execution. The 50 handwritten declarations comprise 20 definitions/structures, 27 theorems, and 3 named instances. `Audit2161.lean` prints the axiom dependencies of all 50; `Check.lean` also displays the definitions and theorem types. The project contains no admitted proof, `native_decide`, or custom axiom. Small finite checks use ordinary kernel-checked `decide`. Builds and source replays treat warnings as errors. An additional inspection inventories generated declarations and checks transitive logical dependencies. Every logical declaration depends only on `propext`, `Classical.choice`, and `Quot.sound`. Compiler runtime artifacts are inventoried separately and are absent from all logical dependency closures.

The printed inequality. The purported universal inequality $\alpha \leq i_+$ is not used as an axiom: for example, P_3 has $\alpha = 2$ and $i_+ = 1$. That observation is only a scope check. The disproof above rests on P_4 , which attains the equality exactly as printed, together with its genus-zero embedding.

5 Independent computations and reproduction

5.1 Exact auxiliary checks

The separate Python checker in `auxiliary/` uses exact rational arithmetic and only the standard library. It enumerates all 16 vertex subsets and expands the determinant by all 24 permutations. For $p(t) = t^4 - 3t^2 + 1$, its exact Sturm sequence is

$$p(t), \quad 4t^3 - 6t, \quad \frac{3}{2}t^2 - 1, \quad \frac{10}{3}t, \quad 1.$$

The sign-variation counts at $-\infty, 0, +\infty$ are 4, 2, 0, giving two negative and two positive roots and no zero root. The checker handles multiplicity through square-free layers; a repeated-root test is included. The Sturm convention is documented in [3].

The same checker independently verifies every dart, vertex rotation and face orbit. It realizes the face as an oriented six-sided polygon with paired sides, reconstructs the quotient vertices, and verifies each vertex link is a circle. It also checks the straight-line drawing with vertices $(0, 0), (1, 0), (2, 0), (3, 0)$ for edge crossings and nonincident vertices in edges. Boundary tests include K_2, P_3, K_3 , and deliberately invalid certificates/drawings. The supplied certificate records all intermediate values, including $(V, E, F) = (4, 3, 1)$ and $g = 0$.

5.2 Commands

From the submission folder, with Python 3 and the pinned Lean toolchain:

```
cd lean
lake exe cache get
lake build
lake env lean -DwarningAsError=true Conjecture2161.lean
lake env lean -DwarningAsError=true Audit2161.lean
lake env lean -DwarningAsError=true Check.lean
cd ..
python3 auxiliary/replay.py
```

The pinned manifest must be preserved. Initial dependency/cache retrieval requires network access. The replay runs the exact checker afresh and requires byte-for-byte agreement with the supplied certificate. `README.md` and `verification/` document the independent clean-build procedure, dependency identities, source hashes, trust audit, and local semantic review. These are local verification records, not maintainer acceptance or a replacement for the repository's review process.

References

- [1] J. L. Gross and T. W. Tucker, *Enumerating Graph Embeddings and Partial-Duals by Genus and Euler Genus*, Enumerative Combinatorics and Applications 1:1 (2021), Article S2S1. [doi:10.54550/ECA2021V1S1S1](https://doi.org/10.54550/ECA2021V1S1S1).
- [2] Y. Nakatsukasa and V. Noferini, *Inertia laws and localization of real eigenvalues for generalized indefinite eigenvalue problems*. [arXiv:1711.00495](https://arxiv.org/abs/1711.00495).
- [3] P. Pébay, J. M. Rojas, and D. C. Thompson, *Sturm's Theorem with Endpoints*. [arXiv:2208.07904](https://arxiv.org/abs/2208.07904).